Deepshikha Chowdhury UG/02/BTCSEAIML/2023/063

UDB LAB 7

Data:

{ roll: 1, name: "Anu", age: 22, department: "IT", marks: 85 },

{ roll: 2, name: "Riya", age: 21, department: "CSE", marks: 78 },

{ roll: 3, name: "Aman", age: 23, department: "IT", marks: 92 },

{ roll: 4, name: "Neha", age: 20, department: "ECE", marks: 74 },

{ roll: 5, name: "Raj", age: 24, department: "CSE", marks: 88 },

{ roll: 6, name: "Simran", age: 22, department: "IT", marks: 81 },

{ roll: 7, name: "Karan", age: 21, department: "ECE", marks: 69 },

{ roll: 8, name: "Priya", age: 23, department: "CSE", marks: 95 }.

```
db.student_UG_02_BTCSEAIML_2023_063.insertMany([
{ roll: 1, name: "Anu", age: 22, department: "IT", marks: 85 },
{ roll: 2, name: "Riya", age: 21, department: "CSE", marks: 78 },
{ roll: 3, name: "Aman", age: 23, department: "IT", marks: 92 },
{ roll: 4, name: "Neha", age: 20, department: "ECE", marks: 74 },
{ roll: 5, name: "Raj", age: 24, department: "CSE", marks: 88 },
{ roll: 6, name: "Simran", age: 22, department: "IT", marks: 81 },
{ roll: 7, name: "Karan", age: 21, department: "ECE", marks: 69 },
{ roll: 8, name: "Priya", age: 23, department: "CSE", marks: 95 }
])
```

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69ca16b903512a77c544152e'),
    '1': ObjectId('69ca16b903512a77c544152f'),
    '2': ObjectId('69ca16b903512a77c5441530'),
    '3': ObjectId('69ca16b903512a77c5441531'),
    '4': ObjectId('69ca16b903512a77c5441532'),
    '5': ObjectId('69ca16b903512a77c5441533'),
    '6': ObjectId('69ca16b903512a77c5441534'),
    '7': ObjectId('69ca16b903512a77c5441535')
  }
}
[
  {
    _id: ObjectId('69c0e639752b6f69fd44152e'),
    roll: 1,
    name: 'Anu',
    age: 22,
    department: 'IT',
    marks: 90,
    status: 'Pass'
  },
  {
    _id: ObjectId('69c0e639752b6f69fd44152f'),
    roll: 2,
    name: 'Riya',
    age: 21,
    department: 'CSE',
```

Q1. Display all documents.

db.student_UG_02_BTCSEAIML_2023_063.find**()**

```
  },
  {
    _id: ObjectId('69c0e639752b6f69fd441530'),
    roll: 3,
    name: 'Aman',
    age: 23,
    department: 'IT',
    marks: 92,
    status: 'Pass'
  },
  {
    _id: ObjectId('69c0e639752b6f69fd441531'),
    roll: 4,
    name: 'Neha',
    age: 20,
    department: 'ECE',
    marks: 74,
    status: 'Pass'
  },
  {
    _id: ObjectId('69c0e639752b6f69fd441532'),
    roll: 5,
    name: 'Raj',
    age: 24,
    department: 'CSE',
    marks: 88,
    status: 'Pass'
  },
```

Q2. Find students from IT department

db.student_UG_02_BTCSEAIML_2023_063.find**({** department**:** "IT" **})**

```
[
  {
    _id: ObjectId('69c0e639752b6f69fd44152e'),
    roll: 1,
    name: 'Anu',
    age: 22,
    department: 'IT',
    marks: 90,
    status: 'Pass'
  },
  {
    _id: ObjectId('69c0e639752b6f69fd441530'),
    roll: 3,
    name: 'Aman',
    age: 23,
    department: 'IT',
    marks: 92,
    status: 'Pass'
  },
  {
    _id: ObjectId('69c0e639752b6f69fd441532'),
    roll: 5,
    name: 'Raj',
    age: 24,
    department: 'CSE',
    marks: 88,
    status: 'Pass'
  },
```

Q3. Find students with marks > 80

db.student_UG_02_BTCSEAIML_2023_063.find**({** marks: **{** $gt: 80 **} })**

```
  {
    _id: ObjectId('69c0e639752b6f69fd441533'),
    roll: 6,
    name: 'Simran',
    age: 22,
    department: 'IT',
    marks: 81,
    status: 'Pass'
  },
  {
    _id: ObjectId('69c0e639752b6f69fd441535'),
    roll: 8,
    name: 'Priya',
    age: 23,
    department: 'CSE',
    marks: 95,
    status: 'Pass'
  },
  {
    _id: ObjectId('69c0e683752b6f69fd441536'),
    roll: 1,
    name: 'Anu',
    age: 22,
    department: 'IT',
    marks: 85,
    status: 'Pass'
  },
```

Q4. Insert one new student

db.student_UG_02_BTCSEAIML_2023_063.insertOne**(**

{ roll: 9, name: "Rahul", age: 22, department: "IT", marks: 83 **}**

)

```
  {
    _id: ObjectId('69c0e685752b6f69fd44153e'),
    roll: 9,
    name: 'Rahul',
    age: 22,
    department: 'IT',
    marks: 83,
    status: 'Pass'
  },
```

Q5. Update marks of Anu to 90.

db.student_UG_02_BTCSEAIML_2023_063.updateOne**(**

{ name: "Anu" **},**

{ $set: **{** marks: 90 **} }**

)

```
{
  _id: ObjectId('69c0e639752b6f69fd44152e'),
  roll: 1,
  name: 'Anu',
  age: 22,
  department: 'IT',
  marks: 90,
  status: 'Pass'
},
```

Q6. Add a new field "status: Pass" for all students

db.student_UG_02_BTCSEAIML_2023_063.updateMany**(**

{},

{ $set: **{** status: "Pass" **} }**

)

```
{
  _id: ObjectId('69c0e639752b6f69fd44152f'),
  roll: 2,
  name: 'Riya',
  age: 21,
  department: 'CSE',
  marks: 78,
  status: 'Pass'
},
{
  _id: ObjectId('69c0e639752b6f69fd441530'),
  roll: 3,
  name: 'Aman',
  age: 23,
  department: 'IT',
  marks: 92,
  status: 'Pass'
},
{
  _id: ObjectId('69c0e639752b6f69fd441531'),
  roll: 4,
  name: 'Neha',
  age: 20,
  department: 'ECE',
  marks: 74,
  status: 'Pass'
},
```

Q7. Delete all students with marks < 70

db.student_UG_02_BTCSEAIML_2023_063.deleteMany**(**

{ marks: **{** $lt: 70 **} }**

)

```
Type "it" for more
{
  acknowledged: true,
  insertedId: ObjectId('69ca16b903512a77c5441536')
}
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 0,
  upsertedCount: 0
}
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 62,
  modifiedCount: 9,
  upsertedCount: 0
}
{ acknowledged: true, deletedCount: 1 }
24
```

Q8. Count students in CSE department

db.student_UG_02_BTCSEAIML_2023_063.countDocuments**(**

{ department**:** "CSE" **}**

)

```
33
```

Q9. Find total students in each department

db.student_UG_02_BTCSEAIML_2023_063.aggregate**([**

{ $group**: {** _id**:** "$department"**,** total**: {** $sum**:** 1 **} } }**

])

```
[
  { _id: 'CSE', total: 33 },
  { _id: 'IT', total: 41 },
  { _id: 'ECE', total: 11 }
]
```

Q10. Find highest marks

db.student_UG_02_BTCSEAIML_2023_063.find**()**

.sort**({** marks**:** -1 **})**

.limit**(**1**)**

```
[
  {
    _id: ObjectId('69c0e639752b6f69fd441535'),
    roll: 8,
    name: 'Priya',
    age: 23,
    department: 'CSE',
    marks: 95,
    status: 'Pass',
    grade: 'A'
  }
]
```

Q11. Find the second highest marks.

db.student_UG_02_BTCSEAIML_2023_063.find()

.sort({ marks: -1 })

.skip(1)

.limit(1)

```
[
  {
    _id: ObjectId('69c0e639752b6f69fd441535'),
    roll: 8,
    name: 'Priya',
    age: 23,
    department: 'CSE',
    marks: 95,
    status: 'Pass',
    grade: 'A'
  }
]
```

Q12. Find students whose name starts with 'A' and marks > 80

db.student_UG_02_BTCSEAIML_2023_063.find({

name: { $regex: "^A" },

marks: { $gt: 80 }

})

```
    grade: 'A'
  },
  {
    _id: ObjectId('69ca188d03512a77c5441540'),
    roll: 1,
    name: 'Anu',
    age: 22,
    department: 'IT',
    marks: 85,
    status: 'Pass',
    grade: 'B'
  },
  {
    _id: ObjectId('69ca188d03512a77c5441542'),
    roll: 3,
    name: 'Aman',
    age: 23,
    department: 'IT',
    marks: 92,
    status: 'Pass',
    grade: 'A'
  }
]
```

Q13. Display department-wise average marks, only for departments where avg > 80

db.student_UG_02_BTCSEAIML_2023_063.aggregate([

{ $group: { _id: "$department", avgMarks: { $avg: "$marks" } } },

{ $match: { avgMarks: { $gt: 80 } } }

])

```
[
  { _id: 'IT', avgMarks: 85.53658536585365 },
  { _id: 'CSE', avgMarks: 87 }
]
```

Q14. Find students with marks between 75 and 90, sorted by marks ascending

db.student_UG_02_BTCSEAIML_2023_063.find(

{ marks: **{** $gte: 75, $lte: 90 **} }**

).sort({ marks: 1 **})**

```
[
  {
    _id: ObjectId('69c0e639752b6f69fd44152f'),
    roll: 2,
    name: 'Riya',
    age: 21,
    department: 'CSE',
    marks: 78,
    status: 'Pass',
    grade: 'C'
  },
  {
    _id: ObjectId('69c0e683752b6f69fd441537'),
    roll: 2,
    name: 'Riya',
    age: 21,
    department: 'CSE',
    marks: 78,
    status: 'Pass',
    grade: 'C'
  },
```

```
{
  _id: ObjectId('69ca16b903512a77c544152f'),
  roll: 2,
  name: 'Riya',
  age: 21,
  department: 'CSE',
  marks: 78,
  status: 'Pass',
  grade: 'C'
},
{
  _id: ObjectId('69c0e6a5752b6f69fd441544'),
  roll: 6,
  name: 'Simran',
  age: 22,
  department: 'IT',
  marks: 81,
  status: 'Pass',
  grade: 'B'
},
```

Q15. Add grade field based on marks

db.student_UG_02_BTCSEAIML_2023_063.updateMany**(**

{},

[

```
{
$set: {
grade: {
$switch: {
branches: [
{ case: { $gte: ["$marks", 90] }, then: "A" },
{ case: { $gte: ["$marks", 80] }, then: "B" },
{ case: { $gte: ["$marks", 70] }, then: "C" }
],
default: "D"
}
}
}
}
]
)
```

```
Type "it" for more
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 85,
  modifiedCount: 0,
  upsertedCount: 0
}
```